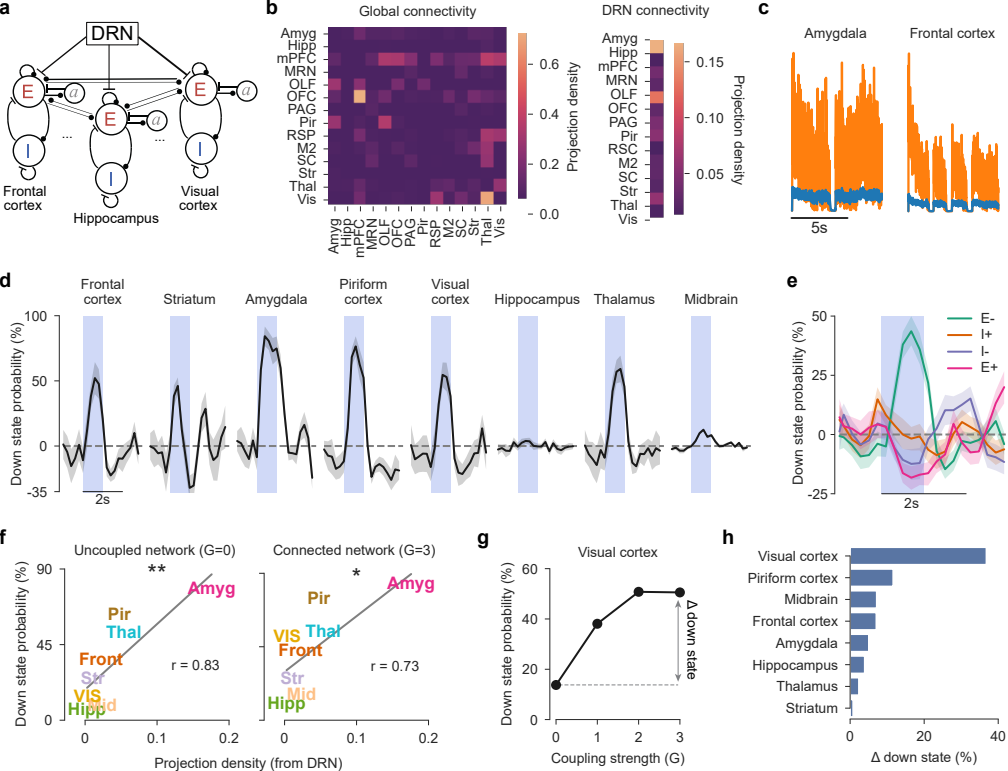

a
DRN
E
I
a
Frontal cortex
Hippocampus
Visual cortex
b
Global connectivity
Amyg
Hipp
mPFC
MRN
OLF
OFC
PAG
Pir
RSP
M2
SC
Str
Thal
Vis
Projection density
0.6
0.4
0.2
0.0
DRN connectivity
0.15
0.10
0.05
c
Amygdala
Frontal cortex
5s
d
Frontal cortex
Striatum
Amygdala
Piriform cortex
Visual cortex
Hippocampus
Thalamus
Midbrain
Down state probability (%)
100
50
0
-35
2s
e
Down state probability (%)
50
25
0
-25
E-
I+
I-
E+
2s
f
Uncoupled network (G=0)
Connected network (G=3)
**
*
Amyg
Pir
Thal
Front
Str
VIS
Hipp
Mid
r = 0.83
r = 0.73
Down state probability (%)
90
45
0
0
0.1
0.2
Projection density (from DRN)
g
Visual cortex
Down state probability (%)
60
40
20
0
Δ down state
0
1
2
3
Coupling strength (G)
h
Visual cortex
Piriform cortex
Midbrain
Frontal cortex
Amygdala
Hippocampus
Thalamus
Striatum
0
20
40
Δ down state (%)